A Benchmark for Evaluating LLM Understanding of Autonomous Swarm Robotics Across Domains

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Abstract

Large Language Models (LLMs) are increasingly used to interpret and reason about complex technical systems, including safety-critical domains such as autonomous swarm robotics. However, despite their strong linguistic capabilities, they often produce hallucinated or overconfident responses, raising concerns about their reliability. In this work, we introduce a text-based benchmark to evaluate LLM understanding of autonomous swarm robotics across six application domains. The dataset consists of 18 structured scenarios and 54 questions targeting technical reasoning, safety awareness, and ethical considerations. We evaluate four LLMs using a multi-dimensional scoring framework covering technical accuracy, reasoning clarity, hallucination, safety awareness, ethical awareness, and overconfidence. We also compare human evaluation with AI-based evaluation. Our results reveal two statistically distinct performance tiers: Claude Sonnet 4.6 and ChatGPT 5.3 Instant produced coherent, well-structured responses with no hallucinations, while Gemini 3 Flash and Perplexity AI Sonar hallucinated in 14.8% and 18.5% of responses. We further find that AI-based scoring with DeepSeek V4 Pro is systematically harsher than human scoring: it rates 95% of responses lower, agrees only weakly with human scores (weighted Cohen's kappa $\kappa = 0.12$), and reverses the human ranking of the top two models, indicating potential bias in automated evaluation. These findings highlight the need for structured benchmarks and human oversight when applying LLMs in safety-critical engineering domains.

1 Introduction

Recent advances in Large Language Models (LLMs) have enabled systems capable of generating fluent explanations of complex technical topics. These models are increasingly used by engineers, researchers, and policymakers to interpret and reason about emerging technologies. However, despite their strong linguistic capabilities, LLMs are known to produce hallucinated information, forget critical constraints, or present overly simplified interpretations. These limitations raise important concerns when LLMs are applied to safety-critical domains.

This work focuses on autonomous swarm robotics, where multiple agents coordinate in dynamic and uncertain environments to perform tasks such as search & rescue, surveillance & security, environmental monitoring, space exploration, agriculture, and underwater exploration. These systems involve complex interactions between perception, communication, and control, and require careful consideration of safety constraints, coordination strategies, and environmental risks. As LLMs are increasingly explored for applications such as planning, analysis, and human-robot interaction, it becomes important to evaluate whether they can accurately interpret and reason about these systems.

While prior work has evaluated LLMs on general reasoning and knowledge tasks, relatively little research has focused on their ability to understand domain-specific, safety-critical engineering systems. In particular, there is a lack of structured benchmarks for assessing how LLMs reason about swarm robotics scenarios across different application contexts.

In this work, we propose a text-based benchmark for evaluating LLM understanding of autonomous swarm robotics across multiple domains. We construct a dataset of realistic scenario descriptions and analytical questions spanning six application areas, and evaluate model responses using a structured scoring framework that measures technical accuracy, safety awareness, hallucination, ethical awareness, reasoning clarity, and overconfidence. We apply this benchmark to four widely used LLMs and perform both quantitative and qualitative analysis of their performance. To the best of our knowledge, this is one of the first benchmarks specifi-

cally designed to evaluate LLM reasoning about autonomous swarm robotics across multiple application domains.

Our contributions are threefold:

- We introduce a domain-specific benchmark for evaluating LLM reasoning in autonomous swarm robotics scenarios across diverse real-world contexts.
- We provide a comparative analysis of four LLMs across 216 responses using a structured, multi-dimensional evaluation framework with statistical significance testing.
- We analyze the reliability of AI-based evaluation by comparing human scoring with automated scoring, highlighting potential biases in AI evaluators.

Our findings show that the evaluated models form two statistically distinct performance tiers, with the weaker tier hallucinating in up to 18.5% of responses. We also find that an LLM-based judge scores responses systematically lower than human evaluators and reverses their ranking of the top two models, suggesting that automated scoring methods may introduce bias. These results highlight both the strengths and limitations of LLMs in interpreting safety-critical robotics systems and underscore the need for careful evaluation and human oversight when deploying generative AI in technical domains.

2 Differences from Proposal

While the overall direction and methodology of the project remained consistent with the original proposal, we made several refinements during implementation to improve the quality and scope of our evaluation.

First, we incorporated an AI-based evaluator (DeepSeek V4 Pro) to complement human scoring. We used this evaluator to assess responses from the top-performing models identified in our analysis, namely ChatGPT 5.3 Instant and Claude Sonnet 4.6, and compared its scores with human evaluations. This allowed us to examine the reliability and potential bias of automated evaluation methods, which was not included in the initial proposal.

Second, we adjusted the set of evaluated models by selecting Perplexity AI Sonar instead of the originally proposed LLaMA-based model. This change was made to ensure accessibility and consistency, as Perplexity AI Sonar provides a stable and easily usable interface for generating responses across all scenarios.

Third, we refined our dataset design by merging the originally separate surveillance and security domains into a single domain, as they exhibited significant conceptual overlap. To maintain diversity in application areas, we introduced a new domain: underwater exploration. This modification allowed us to better capture a wider range of real-world Autonomous Underwater Vehicles (AUVs) and drone swarm robotics scenarios.

Aside from these changes, the core structure of the benchmark remained unchanged. We retained all proposed evaluation metrics, including technical accuracy, reasoning clarity, hallucination, safety awareness, ethical awareness, and overconfidence, and followed the same scenario-based evaluation methodology as outlined in the proposal.

3 Related Work

Our project draws from three main areas of prior work: swarm robotics, LLM-based robotic swarm interaction, and LLM evaluation benchmarks.

Prior work in swarm robotics demonstrates the technical complexity of coordinating multiple autonomous agents in real-world environments. For example, Zhou et al. (2022) show that micro flying robot swarms can navigate outdoor environments using onboard sensing, trajectory planning, obstacle avoidance, and inter-robot coordination. Similarly, work on autonomous drone swarm navigation and multi-target tracking studies how swarms operate in three-dimensional environments with dynamic obstacles (Qamar et al., 2022). These works motivated our dataset design by showing that realistic swarm robotics scenarios must include constraints such as limited sensing, communication, collision avoidance, changing environments, and multi-agent coordination.

Recent research has also explored how LLMs can support swarm robotics. Strobel et al. (2024) propose using LLMs to help robot swarms reason, plan, and collaborate through natural language. Schuck et al. (2024) combine LLMs with safe motion planning for drone swarm choreography, showing the importance of separating high-level language-based planning from low-level safety constraints. Liu et al. (2024) investigate language-guided pattern formation for swarm robotics us-

ing multi-agent reinforcement learning, while Kim et al. (2025) introduce a context-aware multimodal interaction system for human-swarm communication. These works inspired our focus on evaluating whether LLMs can correctly reason about swarm robotics concepts rather than simply generate fluent explanations.

Our benchmark methodology is inspired by broader LLM evaluation research. MMLU evaluates models across many knowledge and reasoning domains, demonstrating the value of structured multi-domain evaluation (Hendrycks et al., 2020). BIG-bench similarly evaluates language models across diverse tasks and highlights that model performance can vary significantly depending on the task type (Srivastava et al., 2022). A broader survey of LLM evaluation emphasizes that models should be assessed not only for task accuracy, but also for reasoning ability, reliability, ethics, and societal risks (Chang et al., 2023). These ideas influenced our decision to evaluate LLM responses using multiple dimensions rather than relying on a single overall score.

Hallucination evaluation is especially relevant to our project because hallucinated information can be dangerous in safety-critical engineering contexts. DefAn evaluates hallucinations in LLM responses using a large-scale benchmark designed to test definitive answers across multiple domains (Rahman et al., 2024). FaithBench studies hallucination in summarization and shows that detecting unsupported model claims remains difficult even for modern evaluation systems (Zhao et al., 2024). These works motivated our decision to include hallucination and overconfidence as separate evaluation rubrics, since a response can appear fluent and confident while still introducing unsupported or misleading claims.

Finally, reports on real-world drone swarm applications highlight the societal relevance of this topic. The U.S. Government Accountability Office describes drone swarms as systems that coordinate multiple drones using algorithms and local sensors with limited human intervention, with possible applications in areas such as wildfire response, monitoring, and defense (U.S. Government Accountability Office, 2023). Work discussing precision agriculture robotics also shows how swarm systems may support food production and environmental sustainability (RoboHub, 2024). These applications influenced our choice of domains, including surveillance and security, agriculture, environmental monitoring, search and rescue, space exploration, and underwater exploration.

Overall, existing work shows that swarm robotics is technically complex, that LLMs are increasingly being explored as interfaces or planners for robotic swarms, and that LLM evaluation requires careful attention to reliability, hallucination, and societal impact. However, there is still limited work evaluating whether general-purpose LLMs can reason accurately about autonomous swarm robotics across different application domains. Our project addresses this gap by constructing a domain-specific benchmark and evaluating model responses using technical, safety, ethical, hallucination, reasoning, and overconfidence-based rubrics.

4 Data

4.1 Dataset Construction

We constructed a text-based benchmark dataset to evaluate LLM understanding of autonomous swarm robotics across multiple real-world application domains. The dataset consists of structured scenario descriptions designed to simulate realistic swarm robotics environments involving multiple agents, dynamic conditions, and safety constraints.

We defined six application domains: surveillance and security, agriculture, environmental monitoring, search and rescue, space exploration, and underwater exploration. Each domain captures distinct challenges such as communication limitations, environmental uncertainty, multi-agent coordination, and risk-sensitive decision-making.

The final dataset contains 18 scenarios, with 3 scenarios per domain. Since each scenario includes three questions, the benchmark contains 54 total questions. Each of the four LLMs was evaluated on all 54 questions, resulting in a total of 216 responses.

4.2 Scenario Design

Each scenario in the dataset includes a detailed description of a swarm robotics setting, specifying factors such as the environment, number of agents, sensing capabilities, communication constraints, and task objectives. The scenarios are designed to reflect realistic operational conditions rather than simplified or idealized settings.

For each scenario, we generated three types of analytical questions:

- **Technical reasoning:** evaluates understanding of swarm coordination, navigation, and

system design.

- **Safety awareness:** evaluates the model's ability to identify risks, constraints, and potential failure modes.
- **Ethical considerations:** evaluates awareness of societal and ethical implications of deploying swarm systems.

This structure ensures that model responses are evaluated across multiple dimensions beyond basic correctness.

4.3 Data Collection and Usage

All scenarios and questions were manually designed based on concepts from swarm robotics literature and real-world applications. We used the same set of scenarios and questions to evaluate all models to ensure consistency and fairness.

Each LLM received the same basic instruction, followed by identical scenario descriptions and questions. The generated responses were then collected and evaluated using our multi-dimensional scoring framework. This controlled setup allows for direct comparison of model performance across domains and evaluation criteria.

5 Methods

5.1 Overview

Our evaluation framework follows a structured pipeline: scenario design, question formulation, response collection, human evaluation, AI-based evaluation, and comparative analysis. This pipeline allows us to systematically assess LLM understanding of swarm robotics across multiple domains and evaluation dimensions.

5.2 Model Evaluation Setup

We evaluated four widely used LLMs: ChatGPT 5.3 Instant, Claude Sonnet 4.6, Gemini 3 Flash, and Perplexity AI Sonar. Each model was prompted with identical scenario descriptions and questions to ensure consistency.

For each of the 54 questions, responses were collected from all four models, resulting in a total of 216 model-generated answers. Before the scenarios, each model received the same basic instruction: that it would be given scenarios and questions, and should answer each one concisely and as well as it could. No external tools or additional domain context were provided beyond this instruction, the scenario descriptions and the questions.

5.3 Evaluation Metrics

Model responses were evaluated using a structured scoring framework consisting of six metrics:

- **Technical Accuracy (0–2):** correctness of swarm robotics concepts and solutions
- **Safety Awareness (0–2):** ability to identify risks and constraints
- **Hallucination (0–1):** absence of unsupported or incorrect claims (1 = none present, 0 = present)
- **Ethical Awareness (0–2):** recognition of societal and ethical implications
- **Reasoning Clarity (0–2):** logical structure and coherence of the explanation
- **Overconfidence (0–1):** absence of unjustified certainty (1 = none present, 0 = present)

Each response received a total score out of 10 based on the sum of these metrics. For hallucination and overconfidence, a score of 1 indicates that the response avoided the issue, while 0 indicates that hallucination or overconfidence was present. All scores were assigned as whole numbers. Technical Accuracy assesses whether the core solution and swarm robotics concepts are correct, while Hallucination separately flags any unsupported or fabricated claims (e.g., nonexistent methods or misapplied technologies), so a response can propose a correct overall approach while still containing a hallucinated detail.

5.4 Human Evaluation

All 216 responses were manually evaluated by both authors together using the defined scoring rubric. This ensured that nuanced aspects such as reasoning quality, safety considerations, and hallucination were assessed with contextual understanding.

5.5 AI-Based Evaluation

To assess the reliability of automated evaluation, we used DeepSeek V4 Pro as an AI-based evaluator. DeepSeek V4 Pro was used to score responses from the top-performing models identified during analysis (ChatGPT 5.3 Instant and Claude Sonnet 4.6). These scores were then compared with human evaluations to identify systematic differences and potential bias in automated scoring.

5.6 Analysis

We conducted five main types of analysis:

- **Overall performance comparison:** distribution of total scores across models
- **Category-level analysis:** comparison across evaluation metrics
- **Human vs AI evaluation:** analysis of differences between human and DeepSeek V4 Pro scores
- **Statistical significance testing:** a Friedman test across all four models, followed by pairwise Wilcoxon signed-rank tests with Holm correction (paired, since every model answered the same 54 questions); human-AI agreement was measured with Pearson correlation as well as quadratic-weighted Cohen's kappa κ
- **Hallucination analysis by domain:** number of hallucinated responses per model in each of the six domains

This multi-level analysis allows us to evaluate not only overall model performance, but also the underlying factors contributing to performance differences and the reliability of evaluation methods.

6 Results

We evaluate model performance across four dimensions: (1) overall performance and statistical significance, (2) category-level performance across evaluation metrics, (3) agreement between human and AI-based scoring, and (4) the distribution of hallucinations across domains. Results are summarized in Table 1 and Figures 1–4.

6.1 Overall Model Performance

Figure 1 shows the distribution of total scores across all evaluated models. We observe clear differences in both central tendency and variance.



Figure 1: Distribution of total scores across models.

Claude Sonnet 4.6 achieves the highest mean score (9.48/10) and the most perfect scores (30 of 54). ChatGPT 5.3 Instant performs similarly (9.39/10) and is the most consistent model, with the lowest standard deviation (0.60 vs. 0.67 for Claude Sonnet 4.6) and no score below 8. Gemini 3 Flash has the lowest mean (8.74) and the widest spread of scores (SD 0.99), while Perplexity AI Sonar scores slightly higher on average (8.89) but also varies considerably (SD 0.92).

A Friedman test confirms that the four models differ significantly ($\chi^2 = 45.7$, $p < 0.001$). Pairwise Wilcoxon signed-rank tests with Holm correction show that Claude Sonnet 4.6 and ChatGPT 5.3 Instant both significantly outperform Gemini 3 Flash and Perplexity AI Sonar (all $p < 0.01$), while the differences within each pair are not significant (Claude Sonnet 4.6 vs. ChatGPT 5.3 Instant: $p = 0.40$; Gemini 3 Flash vs. Perplexity AI Sonar: $p = 0.40$). The models therefore form two statistically distinct performance tiers, with Claude Sonnet 4.6 and ChatGPT 5.3 Instant tied at the top.

6.2 Category-Level Performance

Figure 2 presents average scores across the six evaluation metrics. Technical accuracy remains consistently high across all models, suggesting that most systems are capable of producing responses that align with basic swarm robotics concepts.

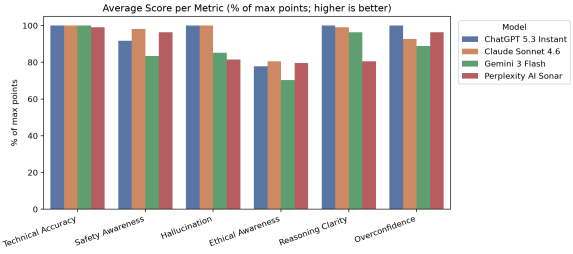


Figure 2: Average score per metric as a percentage of maximum points (higher is better).

Model	Mean (/10)	SD	Perfect 10s	Hallucination rate	Overconfidence rate
Claude Sonnet 4.6	9.48	0.67	30	0.0%	7.4%
ChatGPT 5.3 Instant	9.39	0.60	24	0.0%	0.0%
Perplexity AI Sonar	8.89	0.92	17	18.5%	3.7%
Gemini 3 Flash	8.74	0.99	13	14.8%	11.1%

Table 1: Overall results across 54 questions per model (human scoring). Hallucination and overconfidence rates are the share of responses scored 0 on those metrics.

However, differences emerge in other dimensions. ChatGPT 5.3 Instant and Claude Sonnet 4.6 achieve the highest scores in reasoning clarity, indicating stronger ability to generate coherent and logically structured explanations. Claude Sonnet 4.6 also performs strongly in safety awareness and ethical awareness, suggesting greater attention to risk identification and societal implications.

In contrast, Gemini 3 Flash and Perplexity AI Sonar show weaknesses in different areas. Gemini 3 Flash scores lowest in safety awareness (1.67/2) and ethical awareness (1.41/2), while Perplexity AI Sonar scores lowest in reasoning clarity (1.61/2) despite strong safety awareness (1.93/2). Both models also hallucinated more often: Perplexity AI Sonar in 18.5% of responses and Gemini 3 Flash in 14.8%, compared with no hallucinations for Claude Sonnet 4.6 or ChatGPT 5.3 Instant.

These results suggest that reasoning clarity, safety awareness, and ethical framing are key factors driving differences in overall model performance.

6.3 Human vs AI-Based Evaluation

To assess the reliability of automated evaluation, we compare human scores with scores generated by an AI-based evaluator (DeepSeek V4 Pro), as shown in Figure 3. The same scoring rubric was applied consistently across all responses to reduce evaluation variability.

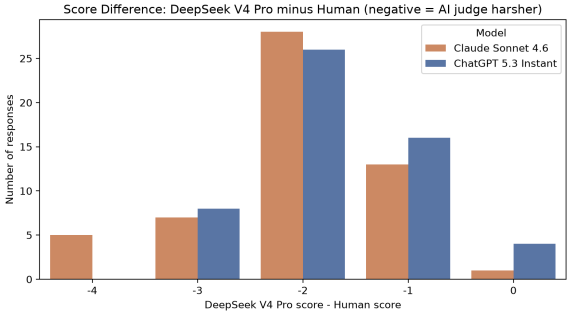


Figure 3: Distribution of score differences (DeepSeek V4 Pro minus human) for Claude Sonnet 4.6 and ChatGPT 5.3 Instant responses.

Across both Claude Sonnet 4.6 and ChatGPT 5.3 Instant responses, DeepSeek V4 Pro consistently assigns lower scores than human evaluators. The distribution of score differences (DeepSeek V4 Pro minus human) is centered below zero, indicating a systematic negative bias. This effect is more pronounced for Claude Sonnet 4.6, where the distribution exhibits greater variance and more extreme negative values.

In contrast, ChatGPT 5.3 Instant shows a narrower distribution of score differences, suggesting more consistent agreement between human and AI-based evaluation, though still with an overall negative shift.

Quantitatively, DeepSeek V4 Pro scored 103 of 108 responses (95%) lower than human evaluators and never scored a response higher, with a mean gap of -1.87 points (-2.04 for Claude Sonnet 4.6 and -1.70 for ChatGPT 5.3 Instant; Wilcoxon signed-rank $p < 0.001$). Agreement between human and AI scores was weak (Pearson $r = 0.49$; quadratic-weighted Cohen's $\kappa = 0.12$). Notably, DeepSeek V4 Pro reversed the human ranking of the two models, scoring ChatGPT 5.3 Instant (7.69) above Claude Sonnet 4.6 (7.44).

These findings indicate that AI-based evaluation may introduce systematic bias and model-dependent variability. As a result, reliance on automated scoring alone may lead to different con-

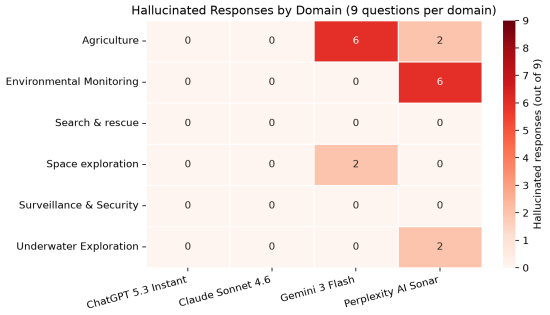


Figure 4: Number of hallucinated responses per model in each domain (9 questions per domain).

clusions about model performance compared to human evaluation.

6.4 Hallucination Patterns Across Domains

Figure 4 shows the number of hallucinated responses per model in each domain. Hallucinations were concentrated rather than evenly spread: 12 of the 18 hallucinated responses came from two model–domain pairs, Gemini 3 Flash in Agriculture (6 of 9 questions) and Perplexity AI Sonar in Environmental Monitoring (6 of 9). Claude Sonnet 4.6 and ChatGPT 5.3 Instant produced no hallucinations in any domain. This suggests that model reliability is domain-dependent, so strong performance in one application area does not guarantee reliability in another.

6.5 Summary of Findings

Across all analyses, four key patterns emerge. First, leading models (Claude Sonnet 4.6 and ChatGPT 5.3 Instant) achieve higher and more consistent performance overall, forming a statistically distinct top tier. Second, differences in reasoning clarity, safety awareness, and ethical framing explain much of the variation in total scores, while technical accuracy alone is not sufficient to distinguish models. Third, AI-based evaluation produces systematically lower scores than human evaluation, with evidence of model-specific bias. Fourth, hallucinations are concentrated in specific model–domain combinations, indicating that model reliability depends on the application domain.

Together, these results provide a comprehensive view of both model capabilities and evaluation reliability within the proposed benchmark.

6.6 Case Study: Gemini 3 Flash Response Analysis

To better understand model behavior, we analyze a representative response generated by Gemini 3 Flash for a swarm robotics scenario.

Scenario: A swarm of 5 drones monitors soil moisture levels across a farm to optimize irrigation. The terrain includes uneven fields and varying crop types. Weather is hot and dry. GPS is reliable, but communication may have slight delays. Drones operate semi-autonomously and coordinate to identify dry zones. Human farmers rely on drone recommendations to control irrigation systems.

Question: How should drones identify and prioritize areas requiring irrigation while minimizing resource usage?

Model Response: To identify and prioritize irrigation needs efficiently, the swarm should utilize multi-modal remote sensing paired with a Weighted Priority Algorithm. The drones use Thermal Infrared (TIR) sensors and Multispectral imaging to calculate the Crop Water Stress Index (CWSI) by comparing canopy temperature to the ambient air temperature; areas with higher temperatures relative to the surroundings are flagged as having low transpiration and high moisture stress. To minimize resource usage, the drones employ hierarchical sampling, where they initially scan the entire field at a high altitude for broad anomalies and then autonomously deploy individual units for low-altitude spot-checks using Soil Moisture Active Passive (SMAP) data or microwave radiometry only in flagged zones. Prioritization is determined by a Scientific Utility Function that weighs the detected moisture deficit against the specific growth stage and "economic value" of the crop in that sector. This results in the generation of a Variable Rate Irrigation (VRI) prescription map, which directs the irrigation system to apply precise water volumes only to the high-stress coordinates, thereby preventing over-saturation of healthy zones and conserving water across the uneven terrain.

Analysis: The response appears fluent and well-structured, but introduces unsupported assumptions about SMAP usage and does not clearly ground the proposed sensing and irrigation strategy in the constraints provided in the scenario, such as drone-level sensing limitations, communication delays, and practical deployment feasibility. This indicates hallucination and overconfidence.

Evaluation: The response scored 0 on both hal-

lucination and overconfidence (total 7/10), as incorrect claims were presented with high certainty.

This example highlights how LLMs may generate convincing yet unreliable explanations in safety-critical contexts.

7 Discussion

7.1 Key Insights

Our findings demonstrate that while modern LLMs exhibit strong linguistic fluency and can generate coherent explanations of swarm robotics systems, their performance varies significantly across evaluation dimensions. In particular, technical accuracy alone does not serve as a reliable indicator of overall model quality. Models that produce technically plausible responses may still fail to account for critical safety constraints, overlook environmental risks, or present incomplete reasoning.

A key observation is that reasoning clarity, safety awareness, and ethical framing are the primary factors distinguishing higher-performing models from lower-performing ones. Models such as Claude Sonnet 4.6 and ChatGPT 5.3 Instant consistently generate more structured and logically coherent responses, while weaker models tend to produce shallow or incomplete analyses. While the top-performing models produced no hallucinations, Claude Sonnet 4.6 still showed overconfidence in 4 of 54 responses, indicating that even strong models can present uncertain claims with unwarranted certainty.

7.2 Societal Implications

The results of this study have important implications for the use of LLMs in safety-critical domains. Swarm robotics systems are increasingly applied in real-world settings such as disaster response, surveillance, and environmental monitoring, where incorrect decisions can have significant consequences. If LLMs are used to support planning, analysis, or human-robot interaction in these domains, errors such as hallucinated constraints or overconfident recommendations could lead to unsafe or suboptimal outcomes.

Our findings highlight the risk of over-reliance on LLM-generated explanations. The fluency and confidence of model outputs may create a false sense of reliability, even when the underlying reasoning is flawed. This is particularly concerning in engineering contexts, where users may not always have the expertise required to detect subtle errors. As a result, human oversight remains essential when deploying LLMs in high-stakes applications.

7.3 Reliability of AI-Based Evaluation

Another important contribution of this work is the analysis of AI-based evaluation. Our comparison between human scoring and automated scoring using DeepSeek V4 Pro reveals a consistent negative bias in AI-generated scores, along with model-dependent variability. This suggests that AI evaluators may apply different implicit criteria than human evaluators, potentially leading to systematic discrepancies in performance assessment.

These findings indicate that while AI-based evaluation can be useful for scaling analysis, it should not be used as a standalone method for benchmarking. Instead, a hybrid approach that combines human judgment with automated evaluation may provide a more reliable and balanced assessment of model performance.

7.4 Limitations

This study has several limitations. First, our dataset is relatively limited in size and scope, and while it covers multiple domains, it may not fully capture the diversity and complexity of real-world swarm robotics scenarios. Second, our evaluation relies on human judgment, which introduces subjectivity and potential bias; because responses were scored jointly by both authors, inter-rater reliability between independent raters could not be measured. Third, the benchmark is text-based and does not incorporate multimodal inputs such as images, sensor data, or simulation environments, which are often critical in robotics applications. Fourth, we evaluate a limited set of LLMs using their publicly accessible versions, which may not reflect the full capabilities of these systems under optimized settings or API-based usage. Finally, the AI-based evaluator was applied only to the two top-performing models.

7.5 Future Work

Future work can extend this benchmark in several directions. Expanding the dataset to include a larger and more diverse set of scenarios would improve the robustness of the evaluation. Incorporating multimodal inputs and simulation-based tasks could provide a more realistic assessment of LLM capabilities in robotics contexts. Furthermore, developing more reliable and interpretable AI-based evaluators remains an open challenge.

Another promising direction is to investigate how LLMs can be integrated with traditional robotics systems in a safe and controlled manner, for example through constrained planning frameworks or human-in-the-loop systems. Such approaches may help mitigate risks associated with hallucination and overconfidence while leveraging the strengths of LLMs in high-level reasoning and communication.

8 Conclusion

This work introduced a benchmark of 54 questions across 18 safety-critical swarm robotics scenarios to evaluate how well LLMs reason about autonomous swarm systems. Across 216 responses from four models, we found two statistically distinct performance tiers: Claude Sonnet 4.6 and ChatGPT 5.3 Instant produced no hallucinations, while Gemini 3 Flash and Perplexity AI Sonar hallucinated in up to 18.5% of responses, with errors concentrated in specific domains. We also found that an LLM-based judge, DeepSeek V4 Pro, scored responses systematically lower than human evaluators and reversed their ranking of the top two models. Together, these results show that while LLMs can generate fluent and structured explanations, they are not yet reliable enough to be used independently in safety-critical decision-making, and that their evaluation should not rely on automated judges alone. Careful benchmarking and continued human oversight remain essential for the responsible integration of LLMs into engineering domains.

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